



UTM
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FACULTY OF BUILT ENVIRONMENT & SURVEYING
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SBEU 3213 FIELD ASTRONOMY
SECTION EB1

Individual Reflection

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“I declare that this reflection is my own work and is based on my personal experiential learning.”

S – Situation

Our experiential learning activity “Beyond the Skies” was an interactive and informative activity targeted towards middle school students. It took place on 12 December 2025 at SMK Taman Universiti. 9 committee members along side with our supervisor, Dr Othman were involved as the representatives of UTM, while 50 form 1 students from SMK Taman Universiti were enlisted as participants.

The activity was structured into several sections. It began with an informative talk introducing the Sun and various celestial objects. This was followed by an interactive quiz session conducted using Kahoot to assess and reinforce the students’ understanding. The second activity was station-based activities, consisting of three stations: a straw rocket workshop, planet exploration, and a crossword puzzle challenge. 50 students are divided into 10 groups for this part of the event. Finally, the programme concluded with a prize-giving ceremony, in which all participants took part in a lucky draw and received a prize.

T – Task

As the Program Director for Beyond the Skies, my primary responsibility was to oversee the overall planning and execution of the programme. Prior to the event, I was required to design the activity flow together with my committee members, delegate tasks according to their roles, and ensure that all preparations were progressing as planned. In addition to this leadership role, I was specifically assigned to be in charge of the Planet Exploration Station, where I served as the game master responsible for guiding students in using the telescope and managing the station activities.

A – Action

To fulfil my role as Program Director, I coordinated planning discussions with the committee, assigned responsibilities clearly and conducted regular follow-ups to monitor progress and address potential issues early. During the event itself, I took an active role at the Planet Exploration Station by demonstrating the correct method of using the telescope and guiding students individually during their observations. I explained the procedures in a fun and engaging manner, including creating a mnemonic chant to help students remember the steps. I also ensured that the activity was hands-on, giving every student at least one opportunity to use the telescope while encouraging their participation and maintaining their excitement throughout the session.

R – Result

As a result of these actions, the programme ran smoothly and the Planet Exploration Station was well received by the students. Four out of ten groups successfully observed the smallest and most challenging sticker placed over 100 metres away, which was a particularly rewarding outcome given the difficulty of the task. Most importantly, all groups were eventually able to observe through the telescope, fulfilling the learning objective of the station. Seeing the students’

fascinated and excited reactions when they successfully made their observations was highly satisfying. This experience strengthened my confidence in leading both a programme and an interactive learning activity, and reinforced the importance of clear guidance, creativity, and hands-on engagement in achieving meaningful learning outcomes.

R – Reflection

Through this experience, I learned that effective leadership requires not only careful planning but also adaptability and active engagement during execution. As the Program Director, I realised that regularly following up with committee members and being present on the ground significantly contributed to the smooth running of the programme. I also learned that I am capable of managing multiple responsibilities simultaneously, balancing overall coordination with hands-on facilitation when needed.

In terms of subject knowledge, this activity reinforced my understanding of astronomy concepts, particularly the **practical application of telescopes**. Explaining telescope usage to secondary school students required me to translate technical procedures into simple, engaging instructions, which aligns with the educational theory that effective learning occurs when complex ideas are delivered at an appropriate cognitive level. This experience also reflected experiential learning principles discussed in class, where learners actively construct understanding through direct participation rather than passive listening.

Working with the students allowed me to better understand the community we were serving. I observed that students were highly curious and motivated when learning was presented in an interactive and enjoyable manner. Their enthusiasm confirmed the **importance of hands-on, inquiry-based activities** in sparking interest in science, particularly for younger students.

If I were to conduct a similar activity in the future, I would allocate more time for practice before the observation task and prepare multiple telescopes to reduce waiting time. I would also consider introducing a short demonstration using visual aids before hands-on practice to further enhance understanding. Overall, this experience deepened my appreciation for experiential learning and strengthened my confidence in leading educational outreach activities.